UNIT 2: SHAPES AND REGIONS

SESSION 2: OBJECT LABELING AND COUNTING

SLO 2: COUNT THE NUMBER OF DISTINCT OBJECTS IN BINARY IMAGES AND EVALUATE THE ACCURACY OF LABELING TECHNIQUES.

Labeling Accuracy

Suppose the actual number of objects = 4 and the algorithm detected = 3 objects.

1. Is the labeling result accurate or inaccurate?

The labeling result is inaccurate because the actual number of objects is 4, but the algorithm detected only 3 objects.

2. Explain the possible reason for the error.

   The possible reason for the error is that one object may have been missed during segmentation or labeling. This can happen due to poor image quality, low contrast between the object and background, noise, overlapping objects, uneven illumination, or an incorrect threshold value. As a result, one of the four objects may not have been detected or may have been incorrectly merged with another object.

Matching

Match the following:

Term	Description
Object Counting	A. Assigning labels to connected regions
Labeling Accuracy	B. Number of separate objects in the image
Connected Component	C. Measure of correctness of labeling
Morphological Processing	D. Helps separate or refine objects

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Reflection Question

Q- Why is accurate object counting important in real-world applications such as traffic monitoring, medical imaging, or industrial inspection?

Accurate object counting is important because it provides correct information about the number of objects present in an image or video. In traffic monitoring, accurate counting of vehicles helps in understanding traffic flow, detecting congestion, and planning better roads and signals. In medical imaging, it helps doctors and medical systems correctly count cells, blood cells, tumors, or other important structures, which can support accurate diagnosis and treatment. In industrial inspection, accurate counting ensures that products, components, or defects are correctly identified, helping maintain quality and reduce manufacturing errors. Therefore, accurate object counting improves decision-making, automation, safety, and overall system reliability in real-world applications.